

MiWay Route Efficiency Analysis

GTFS-Based Route Performance Ranking and Advocacy Priority Framework

Prepared for	MiWay Route Improvement Advocacy
Dataset	MiWay GTFS scheduled service feed; valid May 1, 2026, to June 28, 2026
Representative day	Tuesday, May 5, 2026
Analysis scope	Scheduled service, route geometry, speed, frequency, and route-level scoring
Notebooks	01 GTFS Exploration 02 Geometry 03 Speed 04 Frequency 05 Final Ranking

This analysis converts MiWay's scheduled GTFS feed into a transparent route efficiency ranking and an advocacy priority list. The strongest routes combine high scheduled speed, frequent service, long service span, and balanced bidirectional service. The priority list highlights routes where improvement could matter because the route has meaningful service volume but weaker efficiency indicators.

66 Routes analyzed	5,507 Weekday trips	52 All-day routes	17- 84 Score range
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1. Executive Summary

The project analysed MiWay scheduled service using a GTFS feed valid from May 1 to June 28, 2026. Tuesday, May 5, 2026, was used as the representative service day to avoid mixing weekday and weekend patterns.

Across 66 active routes, the final route efficiency score ranges from 17.46 to 84.14, with a median score of 52.27. The system-level average scheduled speed is 22.12 km/h, and the median route-level headway is 23.75 minutes.

The highest-scoring routes are not simply those with the most trips. They tend to combine speed, all-day coverage, frequent service, and balanced bidirectional service. The advocacy priority lens surfaces routes that may deserve operational attention: substantial service volume with weaker efficiency indicators.

Important: This ranking should be treated as a screening tool. It does not replace ridership, on-time performance, crowding, equity, or rider experience data. A low score is a starting point for investigation, not evidence that a route should be cut.

2. Data Source and Pipeline

The analysis starts from MiWay's standard GTFS feed. GTFS provides a standardized scheduled-service model: routes, trips, stops, stop times, service calendars, and route shapes.

Notebook	Focus	Key Output
01	GTFS exploration & weekday selection	Active trip subset for Tuesday, May 5, 2026
02	Route geometry & stop metrics	Path length, directness ratio, stops per km
03	Scheduled speed & travel time	Trip duration, distance, avg. scheduled speed
04	Frequency & service availability	Headways, service span, directional balance
05	Final efficiency ranking	Component scores, final ranks, advocacy scores

Each notebook produces route-level outputs that feed the next stage. The final scoring notebook joins geometry, speed, and service metrics into a single comparable index.

3. Scoring Methodology

3.1 Component Scores

Each metric is converted to a 0-100 percentile score by ranking routes against each other. For metrics where higher is better, the best route gets the highest score. For metrics where lower is better (headway, stop density), the direction is reversed. Missing values receive a neutral score of 50 so they do not dominate the final ranking.

Component	Weight	Interpretation
Scheduled speed	25%	Higher average scheduled speed scores better.
Frequency	25%	Lower median headway and higher trip volume score better.
Service span	15%	Longer daily span from first to last departure scores better.
Route directness	15%	More direct (non-loop) route geometry scores better.
Stop-spacing	10%	Fewer stops per km scores better as an operating efficiency signal.
Directional balance	10%	More even service between directions scores better.

Final efficiency score formula:

$$= 0.25(\text{speed}) + 0.25(\text{frequency}) + 0.15(\text{service span}) + 0.15(\text{directness}) + 0.10(\text{stop spacing}) + 0.10(\text{directional balance})$$

3.2 Frequency Sub-score

Frequency combines median headway (weighted 70%) and total scheduled trips (weighted 30%). This prevents limited special-purpose routes from ranking highly merely because a few consecutive trips happen to be close together.

3.3 Loop and Circulator Routes

Routes whose endpoint straight-line distance is below 1 km are flagged as loop/circulator routes. Directness can be mathematically misleading for these routes, so they receive a neutral directness score of 50 rather than a penalized score.

Stop-spacing efficiency is an operating efficiency signal, not a service quality verdict. More stops per km can slow a route but stops also provide access. Any stop-consolidation recommendation requires a separate equity and accessibility review.

4. Score Bands and System Distribution

The score bands show that MiWay's scheduled performance is unevenly distributed across the network.

Band	Score Range	Route Count
Very High	Above 85	0
High	70 - 85	7
Moderate	55 - 70	22
Low	40 - 55	17
Very Low	Below 40	20

No routes fall in the very high band. Seven routes achieve a high score, with 22 in the moderate range. The bottom 37 routes split between low and very low, suggesting meaningful headroom for network-wide improvement.

5. Final Route Efficiency Ranking – Top 10

The routes below have the highest final efficiency scores. They generally combine strong scheduled speed, frequent all-day service, and balanced bidirectional coverage.

Rank	Route	Name	Score	Trips	Speed (km/h)	Headway (min)
1	11	Westwood	84.14	172	35.64	15.00
2	35	Eglinton	77.54	159	24.61	15.00
3	1	Dundas	74.65	172	21.84	15.00
4	109	Meadowvale Express	72.72	162	35.02	15.00
5	42	Derry	72.30	171	22.33	15.00
6	66	McLaughlin	71.17	202	20.97	12.00
7	110	University Express	71.15	118	31.64	21.00
8	61	Mavis	69.75	189	21.32	14.00
9	107	Malton Express	69.57	147	26.25	15.00

Rank	Route	Name	Score	Trips	Speed (km/h)	Headway (min)
10	101	Dundas Express	68.90	137	21.46	15.00

Route 11 Westwood leads with a score of 84.14, driven by the highest scheduled speed in the network (35.64 km/h), 15-minute headways, and 172 weekday trips. Route 109 Meadowvale Express is the second-fastest route (35.02 km/h), reflecting its express stop pattern.

6. Advocacy Priority Routes

6.1 How the Priority Score Works

The advocacy priority score is intentionally separate from the efficiency rank. It prioritizes routes with weaker efficiency scores and meaningful service volume, because improvements on those routes may affect more riders.

Priority score formula:

$$= 0.60 \times (100 - \text{efficiency score}) + 0.40 \times \text{service importance score}$$

where service importance is the route's percentile rank by scheduled weekday trips.

6.2 Top 10 Priority Routes

Priority	Route	Name	Priority Score	Efficiency	Trips
1	28	Confederation	68.44	37.96	129
2	6	Credit Woodlands	65.01	29.03	93
3	10	Bristol	64.36	36.16	102
4	2	Hurontario	64.02	59.96	232
5	8	Cawthra	62.99	26.33	71
6	3	Bloor	60.81	62.28	180
7	4	North Service Road	60.37	24.64	55
8	31	Ogden	57.42	45.71	100
9	30	Rexdale	57.28	23.73	48
10	61	Mavis	56.94	69.75	189

The highest-priority review candidates are not necessarily the worst routes - they are routes where the schedule suggests an advocacy conversation could be productive: speed may be low, headways long, stop density high, or directness weak, while the route still carries substantial scheduled service.

Routes 2 (Hurontario), 3 (Bloor), and 61 (Mavis) appear in the advocacy table despite moderate or high efficiency scores because their very high trip volume elevates their service importance, making improvements there more impactful network-wide.

7. Key Findings

- Service availability is broad but uneven. The feed contains 5,507 scheduled weekday trips across 66 active routes; 52 routes meet the all-day service flag.
- Scheduled speed varies sharply. Route-level average speed ranges from 14.95 km/h to 38.17 km/h, a 2.5× spread across the network.
- Frequency is not evenly distributed. The median route headway is 23.75 minutes, but some routes have median headways above 40 minutes.
- Several important corridors, Confederation, Credit Woodlands, Bristol, Hurontario, Cawthra, Bloor, and North Service Road, appear in the advocacy priority lens.
- School routes and special-purpose routes (e.g., Stephen Lewis SS, Streetsville SS) rank low by design. They should be interpreted separately from all-day grid routes.
- No routes currently achieve a Very High (above 85) efficiency score, suggesting network-wide opportunities for improvement.

8. Advocacy Implications

For transit and housing advocacy, the most useful conclusion is not only which route ranks first. The stronger question is where better service would unlock access to housing, campus, jobs, groceries, and late-day travel.

- Use the priority list to identify candidate corridors for deeper public-facing analysis, especially routes with high trip volume and weak efficiency indicators.
- Request ridership, crowding, on-time performance, pass-up, and stop-level boarding data from MiWay and the City of Mississauga to compare scheduled efficiency with lived rider experience.
- Pair route efficiency results with housing density, student housing locations, major employment areas, and late-night travel needs.
- Evaluate whether priority routes need bus priority infrastructure, stop consolidation, branch simplification, schedule restructuring, or better transfer coordination.
- Separate school-special and limited-purpose routes when communicating results publicly their service design goals differ from all-day local routes.
- Consider equity implications before any stop-spacing or frequency recommendations, particularly for routes serving lower-income and transit-dependent communities.

9. Possible Next Steps (If I decide to dig deeper into this analysis)

- Build a sensitivity analysis comparing alternate scoring weights for example, a rider-access-weighted score versus an operating-efficiency-weighted score.
- Add weekend notebooks for Saturday and Sunday to separate commuter, student, retail, and shift-worker travel patterns.
- Compare scheduled speed with real-time GPS or AVL data if vehicle location or historical performance feeds become available from MiWay.
- Produce a short public brief (3-5 pages) with corridor-specific policy asks aimed at City of Mississauga and MiWay decision-makers.
- Revisit the scoring weights with community stakeholders before using the index for formal policy recommendations.

10. Limitations

- The analysis uses scheduled GTFS data only not real-time vehicle movement, actual traffic conditions, or operator behaviour.
- The ranking does not include ridership, crowding, fare equity, physical accessibility barriers, safety, or on-time reliability.
- A single representative weekday keeps the project internally consistent, but weekend and seasonal patterns should be studied separately.
- The final score depends on chosen component weights. Weights should be sensitivity-tested before formal policy use.
- Lower stop density can improve scheduled speed but may harm riders with mobility constraints or those facing longer walks to stops. Any stop-spacing recommendation requires an equity review.
- Loop, school, and limited-purpose routes require separate interpretation because their design goals differ from all-day grid routes.

Figure 1 — MiWay Route Efficiency Score

Graduated colour by final efficiency score (Natural Breaks, 5 classes) — Dark Red = Low, Dark Green = High

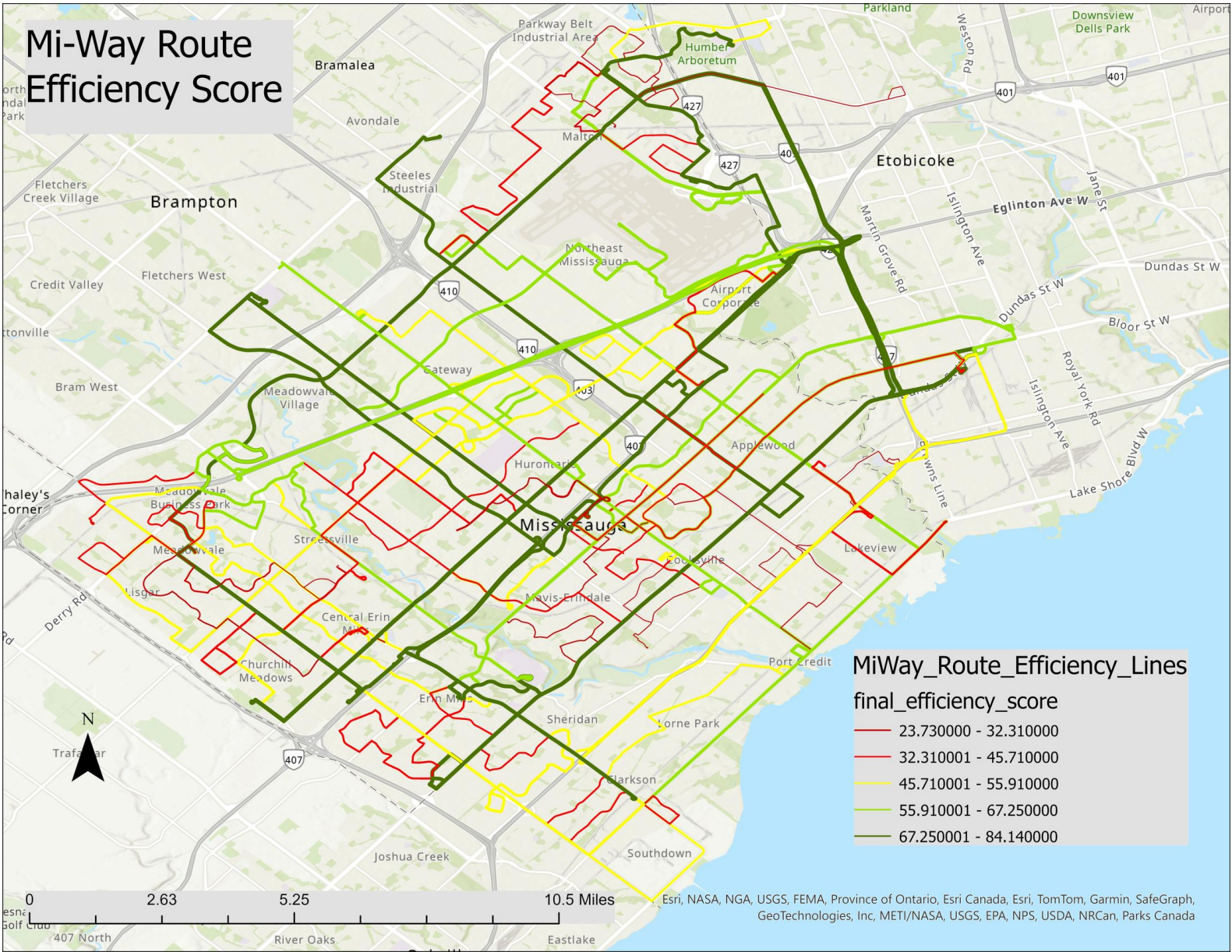
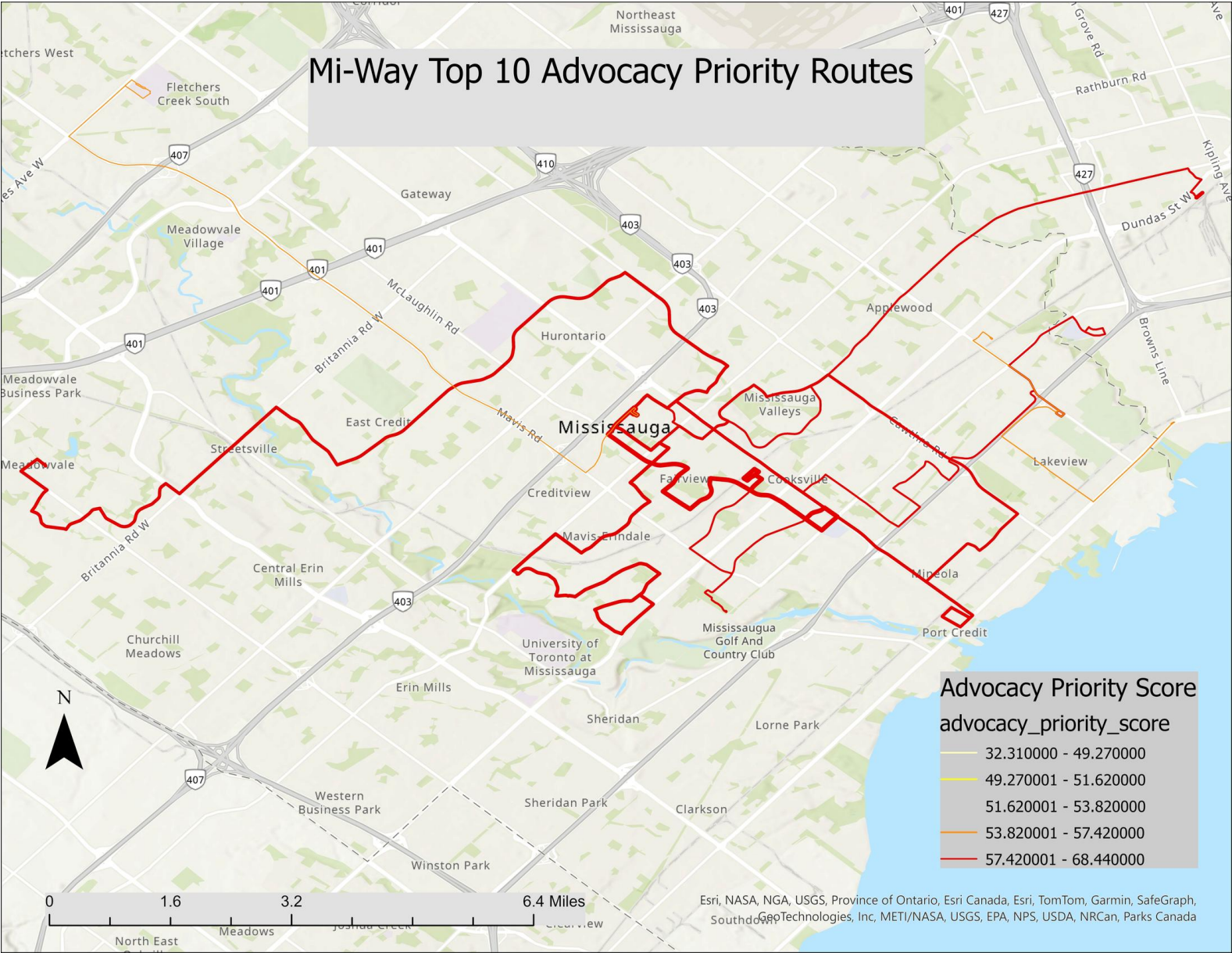


Figure 2 — Top 10 MiWay Advocacy Priority Routes

Routes ranked by advocacy priority score: $0.60 \times \text{improvement need} + 0.40 \times \text{service importance}$



Source: MiWay GTFS scheduled service feed, May 2026. ArcGIS Pro layout export. Analysis: MiWay Route Improvement Advocacy.

Appendix A – Lower Scoring Routes

The routes below have the lowest final scores. Results should be interpreted with service purpose in mind. A school route, loop route, or coverage route may score low while still serving a valid equity or access function.

Rank	Route	Name	Score	Trips	Speed (km/h)	Headway (min)
66	321	Stephen Lewis SS	17.46	5	14.96	33.00
65	30	Rexdale	23.73	48	19.14	55.00
64	4	North Service Road	24.64	55	19.86	44.00
63	313	Streetsville SS	24.68	3	20.39	n/a
62	8	Cawthra	26.33	71	18.60	35.00
61	16	Malton	27.34	50	19.34	40.00
60	6	Credit Woodlands	29.03	93	18.39	28.00
59	315	Rick Hansen SS	32.05	3	14.95	4.00
58	46	Tenth Line	32.31	62	19.73	44.00
57	304	Father Goetz SS	34.63	1	17.43	n/a

Appendix B – Scoring Formula Reference

Frequency sub-score:

$$\text{frequency_score} = 0.70 \times \text{headway_score} + 0.30 \times \text{trip_volume_score}$$

Advocacy priority score:

$$\text{advocacy_priority_score} = 0.60 \times (100 - \text{final_efficiency_score}) + 0.40 \times \text{service_importance_score}$$

Final efficiency score:

$$\text{final_efficiency_score} = 0.25(\text{speed}) + 0.25(\text{frequency}) + 0.15(\text{service span}) + 0.15(\text{directness}) + 0.10(\text{stop spacing}) + 0.10(\text{directional balance})$$

Percentile scoring:

Each raw metric is ranked against other MiWay routes. For higher-is-better metrics, higher percentile = higher score. For lower-is-better metrics (headway, stop density), the percentile is reversed: score = 101 – percentile rank. Missing values receive a neutral score of 50.